

Insurance Charges Regression - Project Report

Raw rows: 1,338 | Clean rows: 1,337 | Duplicate removed: 1

Model Comparison

Model	Train MAE	Test MAE	Test MSE	Test RMSE	Train R2	Test R2	R2 Gap
Decision Tree Regression	2567.47	2621.312	18886631.25	4345.875	0.855	0.897	-0.042
Polynomial Regression	2877.25	2867.317	21585843.724	4646.057	0.834	0.883	-0.049
Linear Regression	4181.902	4177.046	35478020.675	5956.343	0.73	0.807	-0.077

Business Findings

Smokers have average charges of 32,050.23 versus 8,440.66 for non-smokers. Age and BMI are also useful predictive variables, and nonlinear interactions are evaluated through polynomial and tree models. These findings are associations, not causal effects.

Final Model

The lowest held-out test RMSE was achieved by Decision Tree Regression with RMSE 4345.88 and test R-squared 0.897.

Limitations

This educational dataset is limited in scope and representativeness. Real insurance pricing or eligibility use would require actuarial validation, fairness testing, legal review, broader data, monitoring, and external validation.

